Matching and vertex-cover lower bounds for short simple deletion streams via hidden augmented bi-index instances

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Editorial note. Editable transcription of completeformallypublishedCCBY3.0Unported one-way deletionlowerbound theme. Nonexclusive nativev1 privateaid; publicselectedfiles onlycorrespondingformalmodels/Sections2–4 throughlastlowerboundcorollary. FullAugmentedBiIndex, bothgraphreductions anderror/parallelrunanalyses retained together; authorcalls themsame method, noobject/parameter/supportsplit. Alloriginalauthorconstants, inequalities andprobabilityexpressions preserved withoutmathematicalrepair ornewassumptions. AllnecessaryTikZ diagrams preservededitable; omittedintroFigure1 accounted byfigurecounter1. Originalglobaltheoremnumbers restored. Latervertexcoverupperboundsketch omitted. Noinput/include/assets. Allproduction/validationYupeng; rootownscompilation.

Allowed standard prerequisites (editorial).

- Randomized one-way AugmentedIndex lowerbound, bipartite matching/vertexcover and standardconcentration bounds. FullSection2 AugmentedBiIndex sourcelocal reduction retained; standardAugmentedIndex theorem cited, completeSections3/4 graphreductions anderroranalysis included

One-way deletion communication and dynamic graph streams

In our model, Alice holds a set of edges and sends a single message to Bob. Bob holds a set of edge deletions, which form a subset of Alice's edges, and needs to report a large matching or a small vertex cover in the graph spanned by the edges that are not deleted. The lower bound constructions yield insertion-deletion streams of length $O(n^2)$ and do not involve multi-edges. The lower bounds therefore also hold for streaming algorithms that are designed for short input streams and simple graphs. Algorithms may output non-existing edges when they err. A standard reduction shows that a lower bound on the size of Alice's message also constitutes a lower bound on the space requirements of an insertion-deletion streaming algorithm. The primary communication reductions and their stream consequences are retained together below.

2 Augmented Bi-Index

In this section, we define the one-way two-party communication problem Augmented Bi-Index and prove a lower bound on its communication complexity.

Problem 1 (Augmented Bi-Index)**.** *In an instance of* Augmented Bi-Index $\mathsf{BInd}_\delta^{n,k}$ *we have two players denoted Alice and Bob. Alice holds a binary matrix* $A \in \{0,1\}^{n\times n}$*. Bob holds indices* $x, y \in [n-k]$ *and the incomplete*[1] *binary matrix* $A_{S(x,y)}$ *where*

$$S(x,y) = \{(i,j) \in [n]^2 \mid (x \le i < x+k) \text{ and } (y \le j < y+k)\} \setminus \{(x,y)\} \ .$$

Alice sends a single message M *to Bob who must output* $A_{x,y}$ *with probability at least* $1-\delta$*.*

Our lower bound proof consists of a reduction from the well-known Augmented Index problem, which is known to have large communication complexity.

Problem 2 (Augmented Index)**.** *In an instance of* Augmented Index Ind_δ^n *we have two players denoted Alice and Bob. Alice holds a binary vector* $V \in \{0,1\}^n$*. Bob holds an index* $\ell \in [n]$ *and the vector suffix* $V_{>\ell} = (V_{\ell+1}, V_{\ell+2}, \cdots, V_n)$*. Alice sends a single message* M *to Bob who must output* V_ℓ *with probability at least* $1-\delta$*.*

As a consequence of Lemma 13 in [2], we can see that this problem has linear communication complexity (see also Lemma 2 in [1] for a more direct proof technique).

Theorem 3 (e.g. [2])**.** *For* $\delta < 1/3$*, any randomised one-way communication protocol which solves* Ind_δ^n *must communicate* $\Omega(n)$ *bits.*

We are now ready to prove our lower bound for Augmented Bi-Index.

Theorem 4. *For* $\delta < 1/3$*, any randomised one-way communication protocol which solves* $\mathsf{BInd}_\delta^{n,k}$ *must communicate* $\Omega((n-k)^2)$ *bits.*

Proof. Let $\mathcal{P}$ be a communication protocol for $\mathsf{BInd}_\delta^{n,k}$ that uses messages of length at most $S(n,k)$ bits. We will show how $\mathcal{P}$ can be used to solve $\mathsf{Ind}_\delta^{(n-k)^2}$ with the same message size.

Let V, ℓ be any instance of $\mathsf{Ind}_\delta^{(n-k)^2}$. Alice builds the matrix $A \in \{0,1\}^{n\times n}$ by placing the bits of V in lexicographical order in the top-left $(n-k)$-by-$(n-k)$ region:

$$A_{i,j} = \begin{cases} V_{j+(n-k)(i-1)} & \text{for } i,j \in [n-k] \\ 0 & \text{otherwise} \end{cases} \ .$$

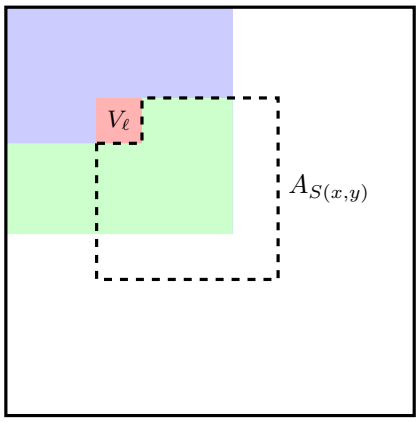
This packing is illustrated in Figure 2(a).

Alice runs protocol $\mathcal{P}$ on A and sends the resulting message M to Bob. Now, Bob has the message M, the index $\ell \in [(n-k)^2]$ and the suffix $V_{>\ell}$. Let $x, y \in [n-k]$ be the unique pair of integers such that $\ell = y + (n-k)(x-1)$. Observe that $A_{x,y} = V_\ell$.

[1] We use A_S to refer to the collection of entries indexed by the set S, so $A_S = (A_{i,j})_{(i,j)\in S}$.

V_1	V_2	V_3	V_4	V_5	0	0	0	0
V_6	V_7	V_8	V_9	V_{10}	0	0	0	0
V_{11}	V_{12}	V_{13}	V_{14}	V_{15}	0	0	0	0
V_{16}	V_{17}	V_{18}	V_{19}	V_{20}	0	0	0	0
V_{21}	V_{22}	V_{23}	V_{24}	V_{25}	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

(a) Example packing of the bits of V into matrix A with $n = 9$ and $k = 4$.



(b) Bob can construct the area $A_{S(x,y)}$ given $V_{>\ell}$, which is part of his input.

Figure 2: The construction of A and $A_{S(x,y)}$ in Theorem 4.

For Bob to be able to complete protocol $\mathcal{P}$ he needs to provide $A_{S(x,y)}$. Because of the way we packed the entries of V onto A, the overlap between V and $A_{S(x,y)}$ is a subset of the entries of $V_{>\ell}$ (see Figure 2(b) for an illustration). Therefore Bob can complete the protocol and determine $A_{x,y} = V_\ell$ with probability at least $1 - \delta$. By Theorem 3, it must be that $S(n, k) = \Omega((n - k)^2)$. □

3 Maximum Matching

Let $\mathbf{A}$ be a C-approximation insertion-deletion streaming algorithm for Maximum Matching that errs with probability at most $1/10$. We will now show that $\mathbf{A}$ can be used to solve $\mathsf{Blnd}_\delta^{n,k}$.

3.1 Reduction

Let $A \in \{0,1\}^{n\times n}, x \in [n-k]$ and $y \in [n-k]$ be an instance of $\mathsf{Blnd}_\delta^{n,k}$. Alice and Bob first sample a uniform random binary matrix $X \in \{0,1\}^{n\times n}$ and random permutations $\sigma_1, \sigma_2 : [n] \to [n]$ from public randomness. Alice then computes matrix A' which is obtained by first computing the entry-wise XOR of A and X, denoted by $A \oplus X$, and then by permuting the rows and columns of the resulting matrix by σ_1 and σ_2, respectively. Next, Alice interprets A' as the incidence matrix of a bipartite graph $G(A')$. Alice runs algorithm $\mathbf{A}$ on a random ordering of the edges of $G(A')$ and sends the resulting memory state to Bob.

Next, Bob also computes the entry-wise XOR between the part of the matrix A that he knows about, $A_{S(x,y)}$, and X, followed by applying the permutations σ_1 and σ_2. In doing so, Bob knows the matrix entries of A' at positions $(\sigma_1(i), \sigma_2(j))$ for every $(i, j) \in S(x, y)$. He can therefore compute the subset E_S of the edges of $G(A')$ with

$$E_S = \{(\sigma_1(i), \sigma_2(j)) \in [n]^2 \mid (i,j) \in S(x,y) \text{ and } A'(\sigma_1(i), \sigma_2(j)) = 1\} \ .$$

Furthermore, let $E_{diag} \subseteq E_S$ be the set of edges $(\sigma_1(i), \sigma_2(j))$ so that (i,j) lies on the same diagonal in A as (x,y), or, in other words, there exists an integer $1 \leq q \leq k-1$ such that $(x+q, y+q) = (i,j)$. Then, let $E_{del} = E_S \setminus E_{diag}$. Bob continues the execution of algorithm $\mathbf{A}$, as follows: for every edge $e \in E_{del}$, Bob introduces an edge deletion of e, in random order.

Let M' be the matching returned by $\mathbf{A}$. From M' Bob computes the matching M as follows: If $|M' \leq 0.99\frac{k}{2C}|$ then Bob sets $M = \varnothing$. Otherwise, Bob sets M to be a uniform random subset of M' of size exactly $0.99\frac{k}{2C}$.

Parallel Executions. Alice and Bob execute the previous process $\ell = 100 \cdot C$ times in parallel. Let M^i, X^i, σ_1^i and σ_2^i be M, X, σ_1 and σ_2 that are used in run i, respectively. Let Q_i be the indicator random variable that is 1 iff M^i contains the edge $(\sigma_1^i(x), \sigma_2^i(y))$. We also define $p = \sum_i Q_i$ to be the total number of times the edges $(\sigma_1^i(x), \sigma_2^i(y))$ are reported. Whenever the edge $(\sigma_1^i(x), \sigma_2^i(y))$ is reported, we interpret this to be a claim that $A_{x,y} = \neg X^i_{x,y}$. So depending on the value of $X^i_{x,y}$, this acts as a claim that $A_{x,y} = 0$ or $A_{x,y} = 1$. We define $p_0 = \sum_{i:Q_i=1} X^i_{x,y}$ (which counts how often $A_{x,y} = 0$ was claimed) and let $p_1 = p - p_0$ (the number of times $A_{x,y} = 1$ was claimed). Bob outputs 1 as his estimator for $A_{x,y}$ if $p_1 \geq p_0$ and 0 otherwise.

3.2 Analysis

Let G be the bipartite graph with incidence matrix $A \oplus X$, and let

$$F = \{(i,j) \in S(x,y) \mid (A \oplus X)_{i,j} = 1 \text{ and } \nexists\ q \text{ s.t. } (i,j) = (x+q, y+q)\} \ .$$

Then the graph $G - F$ is isomorphic to the graph $G(A') - E_{del}$. In particular, $G(A') - E_{del}$ is obtained from $G - F$ by relabeling the vertex sets of the two bipartitions using the permutations σ_1 and σ_2.

We will first bound the maximum matching size in $G(A') - E_{del}$. To this end, we will bound the maximum matching size in $G - F$, which is easier to do:

Lemma 5. *With probability $1 - \frac{1}{k^{10}}$, the graph $G(A') - E_{del}$ is such that:*

$$0.99\frac{k}{2} \leq \mu(G(A') - E_{del}) \leq 1.01\frac{k}{2} + 2(n-k) \ ,$$

where $\mu(G)$ denotes the matching number *of G, i.e., the size of a maximum matching.*

Proof. We will consider the graph $G - F$ instead, since it is isomorphic to $G(A') - E_{del}$ and has the same maximum matching size.

First, observe that G is a random bipartite graph where every edge is included with probability $\frac{1}{2}$. Let U and V denote the bipartitions in G, and consider the subsets $U' = [x, x+k)$ and $V' = [y, y+k)$. Observe that in the vertex induced subgraph $G[U' \cup V']$ all edges are deleted in F except those that connect the vertices $x+i$ and $y+i$, for every $0 \leq i \leq k-1$. By a Chernoff bound, the number of edges and thus the maximum matching size in $G[U' \cup V']$ is bounded by:

$$0.99 \cdot \frac{k}{2} \leq \mu(G[U' \cup V']) \leq 1.01 \cdot \frac{k}{2} \ ,$$

with probability $1 - \frac{1}{k^{10}}$.

Observe that, with probability $1 - \frac{1}{k^{10}}$, the neighborhood $\Gamma(U')$ is such that

$$0.99 \cdot \frac{k}{2} \leq |\Gamma(U')| \leq 1.01 \cdot \frac{k}{2} + (n-k) \ .$$

The set U' can therefore be matched to at most $1.01 \cdot \frac{k}{2} + (n-k)$ vertices in V. We thus obtain

$$\mu(G - F) \leq 1.01 \cdot \frac{k}{2} + 2(n-k) \ ,$$

since we may also be able to match all $n - k$ vertices of $U \setminus U'$. □

Lemma 6. *Suppose that $M_i \neq \varnothing$. Then:*

$$\frac{0.99}{2C} - \frac{2(n-k)}{k} \quad \leq \quad \mathbb{P}\left[Q_i = 1\right] \leq \frac{0.99}{2C} \ .$$

Proof. First, by construction of our reduction, since $M_i \neq \varnothing$ we have $|M_i| = 0.99\frac{k}{2C}$. Let

$$U'_i = \sigma_1^i([x, x+k)) \text{ and } V'_i = \sigma_2^i([y, y+k)) \ .$$

Let $\tilde{M}_i$ be the set of edges of M_i connecting vertices in U'_i to V'_i. Observe that there are $2(n-k)$ vertices in the graph outside the set $U'_i \cup V'_i$. We thus have

$$|M_i| - 2(n-k) \leq |\tilde{M}_i| \leq |M_i| \ .$$

Next, since the permutations σ_1^i, σ_2^i are chosen uniformly at random, any edge of $\tilde{M}_i$ may have originated from any of the diagonal entries in $A_{S(x,y)}$. Hence, $\tilde{M}_i$ claims the bits of at least $|M_i| - 2(n-k)$ and at most $|M_i|$ uniform random positions in the diagonal of $A_{S(x,y)}$. Every entry in the diagonal of $A_{S(x,y)}$ is thus claimed with the same probability. Since the diagonal of $A_{S(x,y)}$ is of length k, this probability is at least

$$\frac{|M_i| - 2(n-k)}{k} = \frac{0.99\frac{k}{2C} - 2(n-k)}{k} = \frac{0.99}{2C} - \frac{2(n-k)}{k} \ ,$$

and at most

$$\frac{|M_i|}{k} = \frac{0.99\frac{k}{2C}}{k} = \frac{0.99}{2C} \ .$$

□

Theorem 7. *Let $\mathbf{A}$ be a n^ϵ-approximation insertion-deletion streaming algorithm for Maximum Matching that errs with probability at most $1/10$ and uses space s. Then there exists a communication protocol for $\mathsf{BInd}_{0.05}^{n, n-\frac{1}{40}n^{1-\epsilon}}$ that communication $\mathrm{O}(n^\epsilon \cdot s)$ bits.*

Proof. Let $C = n^\epsilon$ and let $k = n - \frac{1}{40}n^{1-\epsilon}$. First, by Lemma 5, with probability $1 - \frac{1}{k^{10}}$, the graph $G(A') - E_{del}$ contains a matching of size at least $0.99k/2$. By a union bound, the probability that this graph is of at least this size in each of the ℓ iterations is at least $1 - \frac{\ell}{k^{10}}$. Suppose from now on that this event happens.

Let ℓ_1 be the number of times the algorithm $\mathbf{A}$ succeeds, and let ℓ_0 be the number of times $\mathbf{A}$ errs. Then, $\ell = \ell_0 + \ell_1$. Whenever $\mathbf{A}$ succeeds, since $\mathbf{A}$ is a C-approximation algorithm, the matching M'_i is of size $0.99\frac{k}{2C}$, which further

implies that M_i is of size exactly $0.99\frac{k}{2C}$. Since the algorithm must return a correct matching, every time we have a claim (i.e. $Q_i = 1$), the claimed bit value must be correct. Thus, by Lemma 6, we get a correct claim on $A_{x,y}$ with probability at least

$$\frac{0.99}{2C} - \frac{2(n-k)}{k} = \frac{0.99}{2n^\epsilon} - \frac{2(\frac{1}{40}n^{1-\epsilon})}{n - \frac{1}{40}n^{1-\epsilon}} \geq \frac{0.99}{2n^\epsilon} - \frac{\frac{1}{40}n^{1-\epsilon}}{n} = \frac{0.99}{2n^\epsilon} - \frac{1}{40n^\epsilon} \geq \frac{2}{5n^\epsilon} ,$$

where we used the inequality $\frac{2x}{y-x} \geq \frac{x}{y}$, which holds for every $y > x$. We thus expect to see the correct bit claimed at least $\ell_1 \cdot \frac{2}{5n^\epsilon}$ times in total. On the other hand, incorrect claims of the bit value can only occur when the algorithm errs. In the worst case, **A** will make as many false claims as possible - so we assume the algorithm never results in $M_i = \emptyset$ when it errs. Lemma 6 also allows us to bound the probability of an incorrect claim for this bad algorithm by $\frac{0.99}{2n^\epsilon}$. We thus expect to see the wrong bit value claimed at most $\ell_0 \cdot \frac{0.99}{2C} \leq \frac{\ell_0}{2n^\epsilon}$ times.

Recall that $\ell = 100n^\epsilon$. Then, by standard concentration bounds, the probability that $\ell_0 \geq 2 \cdot \frac{\ell}{10}$ is at most $\frac{1}{100}$ (recall that the error probability of **A** is at most $\frac{1}{10}$). Suppose now that $\ell_0 \leq \frac{1}{5}\ell$ holds, which also implies that $\ell_1 \geq \frac{4}{5}\ell$. We thus expect to learn the correct bit at least

$$\frac{4}{5}100n^\epsilon \cdot \frac{2}{5n^\epsilon} = 32$$

times, and using a Chernoff bound, it can be seen that the probability that we learn the correct bit less than 21 times is at most 0.02. Similarly, we expect to learn the incorrect bit at most

$$\frac{1}{5}100n^\epsilon \cdot \frac{1}{2n^\epsilon} = 10$$

times, and by a Chernoff bound, it can be seen that the probability that we learn the incorrect bit at least 20 times is at most 0.01. Our algorithm therefore succeeds if all these events happen. Taking a union bound over all failure probabilities that occurred in this proof, we see that our algorithm succeeds with probability

$$1 - \frac{100n^\epsilon}{k^{10}} - 0.01 - 0.02 - 0.01 \geq 0.95 .$$

□

Since by Theorem 4, $\mathsf{Blnd}_{0.05}^{n,n-\frac{1}{40}n^{1-\epsilon}}$ has randomized one-way communication complexity $\Omega(n^{2-2\epsilon})$, by Theorem 7 we obtain our main result of this section:

Corollary 8. *Every insertion-deletion n^ϵ-approximation streaming algorithm for Maximum Matching that errs with probability at most $\frac{1}{10}$ requires space $\Omega(n^{2-3\epsilon})$.*

4 Minimum Vertex Cover

Let **B** be a C-approximation insertion-deletion streaming algorithm for Minimum Vertex Cover that succeeds with probability $1 - 1/400$. Similar to the previous section, we will now show how **B** can be used to solve $\mathsf{Blnd}_\delta^{n,k}$.

4.1 Reduction

Let $A \in \{0,1\}^{n\times n}, x \in [n-k]$ and $y \in [n-k]$ be an instance of $\mathsf{BInd}_\delta^{n,k}$. The reduction for Minimum Vertex Cover is very similar to the reduction for Maximum Matching presented in the previous section. Alice's behaviour is in fact identical:

First, Alice and Bob sample a uniform random binary matrix $X \in \{0,1\}^{n\times n}$ and random permutations $\sigma_1, \sigma_2 : [n] \to [n]$ from public randomness. Alice then computes matrix A' which is obtained by first computing $A \oplus X$ and then permuting the rows and then the columns of the resulting matrix by σ_1 and σ_2, respectively. Alice interprets A' as the incidence matrix of a bipartite graph $G(A')$. Alice then runs algorithm $\mathbf{B}$ on a random ordering of the edges of $G(A')$ and sends the resulting memory state to Bob.

Next, Bob also computes the entry-wise XOR between the part of the matrix A that he knows about and X, followed by applying the permutations σ_1 and σ_2. In doing so, Bob knows the matrix entries of A' at positions $(\sigma_1(i), \sigma_2(j))$ for every $(i,j) \in S(x,y)$. He can therefore compute the subset E_S of the edges of $G(A')$ with

$$E_S = \{(\sigma_1(i), \sigma_2(j)) \in [n]^2 \mid (i,j) \in S(x,y) \text{ and } A'(\sigma_1(i), \sigma_2(j)) = 1\} .$$

Next, Bob continues the execution of $\mathbf{B}$ and introduces deletions *for all edges in* E_S in random order. Observe that this step is different to the reduction for Maximum Matching. Let I be the vertex cover produced by $\mathbf{B}$.

Parallel Executions. Alice and Bob run the procedure above 40 times in parallel. Denote by I^i, X^i, E_S^i, A'^i, σ_1^i, and σ_2^i the variables I, X, E_S, A', σ_1 and σ_2 used in iteration i. Furthermore, let Q_i be the indicator variable that is 1 iff $\{\sigma_1^i(x), \sigma_2^i(y)\} \cap I_i \neq \varnothing$, i.e., the potential edge $(\sigma_1^i(x), \sigma_2^i(y))$ is covered by the vertex cover.

If there exists a run j with $Q_j = 0$, then Bob predicts $A_{x,y} = X_{x,y}$ (if there are multiple such runs then Bob breaks ties arbitrarily). Otherwise, Bob returns `fail` and the algorithm errs.

4.2 Analysis

The first lemma applies to every parallel run j. For simplicity of notation, we will omit the superscripts that indicate the parallel run in our random variables.

We first show an upper bound on the size of a minimum vertex cover in $G(A') - E_S$.

Lemma 9. *The size of a minimum vertex cover in* $G(A') - E_S$ *is at most* $2(n-k)+1$.

Proof. Let U, V be the bipartitions of the graph $G(A') - E_S$, let $U' = \{\sigma_1(a) : a \in [x, x+k)\}$ and let $V' = \{\sigma_2(b) : b \in [y, y+k)\}$. Observe that $(G(A') - E_S)[U' \cup V']$ contains at most one edge: The potential edge between $\sigma_1(x)$ and $\sigma_2(y)$. A valid vertex cover of $G(A') - E_S$ is therefore $(U \setminus U') \cup (V \setminus V') + \sigma_1(x)$, which is of size $2(n-k)+1$. □

Next, we prove the key property of our reduction: We show that if $A'_{\sigma_1(x),\sigma_2(y)} = 0$ (or equivalently, $A_{x,y} \oplus X_{x,y} = 0$) then neither $\sigma_1(x)$ nor $\sigma_2(y)$ is in the output vertex cover with large probability.

Lemma 10. *Assume that algorithm* **B** *does not err in run* j*. Suppose that* $A'^{j}_{\sigma_1^j(x),\sigma_2^j(y)} = 0$*. Then the probability that* $Q_j = 1$ *is at most*

$$\frac{3C \cdot (2(n-k)+1)}{k} \ .$$

Proof. Consider the set $D = \{(\sigma_1^j(x+i), \sigma_2^j(y+i)) \mid 0 \leq i \leq k-1\}$, i.e., the positions of the diagonal of $S(x,y) \cup \{x,y\}$ permuted by σ_1^j and σ_2^j. Then, since A'^j is a uniform random matrix, with probability at least $1 - \frac{1}{k^{10}}$, the "permuted diagonal" A'^j_D contains at least $0.99k/2$ entries with value 0, or, in other words, graph $G(A'^j) - E_S^j$ contains at least $0.99k/2$ non-edges in the positions of the permuted diagonal D. By Lemma 9, the size of a minimum vertex cover in $G(A'^j) - E_S^j$ is at most $2(n-k)+1$, and since **B** has an approximation factor of C, the vertex cover I_j is of size at most $C \cdot (2(n-k)+1)$. Hence, at most $C \cdot (2(n-k)+1)$ non-edges in D can be covered in I_j. However, since the permutations are random, the probability that the non-edge $(\sigma_1^j(x), \sigma_2^j(y))$ is covered, which is identical to the event $Q_j = 1$, is therefore at most

$$\frac{C \cdot (2(n-k)+1)}{0.99k/2} \leq \frac{3C \cdot (2(n-k)+1)}{k} \ .$$

□

Theorem 11. *Let* **B** *be a* n^ϵ*-approximation insertion-deletion streaming algorithm for* Minimum Vertex Cover *that uses space* s *and errs with probability at most* $1/400$*. Then, there exists a communication protocol for* $\mathsf{BInd}^{n, n-\frac{1}{20}n^{1-\epsilon}}_{\frac{1}{3}}$ *that communicates* $\mathrm{O}(s)$ *bits.*

Proof. Let $k = n - \frac{1}{40}n^{1-\epsilon}$ and let $C = n^\epsilon$. Consider the reduction given in the previous subsection. First, observe that since **B** errs with probability at most $1/400$, by the union bound the probability that **B** errs at least once in the 40 parallel executions of our reduction is at most $\frac{1}{10}$. We assume from now on that the algorithm never errs.

Observe that the matrices A'^j are random matrices. Hence, the probability that there exists at least one run i with $A'^i_{\sigma_1^i(x),\sigma_2^i(y)} = 0$ is at least $1 - (\frac{1}{2})^{40}$. Suppose that this event happens. Let run i be so that $A'^i_{\sigma_1^i(x),\sigma_2^i(y)} = 0$. Then, by Lemma 10, the probability that the non-edge $(\sigma_1^i(x), \sigma_2^i(y))$ is covered by I_i, or in other words, the probability that $Q_i = 1$, is at most

$$\frac{3C \cdot (2(n-k)+1)}{k} = \frac{3n^\epsilon \cdot (\frac{1}{20}n^{1-\epsilon}+1)}{n - \frac{1}{40}n^{1-\epsilon}} = \frac{\frac{3}{20}n + 3n^\epsilon}{n - \frac{1}{40}n^{1-\epsilon}} = \frac{3}{20} + o(1) \ .$$

Observe that whenever $Q_i = 0$, the algorithm outputs $X^i_{x,y}$ as a predictor for $A_{x,y}$. Since the algorithm **B** does not err, we have $A_{x,y} \oplus X^i_{x,y} = 0$. This implies that $A_{x,y} = X^i_{x,y}$, which establishes correctness.

Last, we need to bound the error probability of our algorithm. First, the probability that at least one of the 40 runs fails is at most $\frac{1}{10}$. Next, the probability that none of the runs are such that $A'^j_{\sigma_1^j(x),\sigma_2^j(y)} = 0$ is at most $(\frac{1}{2})^{40}$.

Furthermore, the probability that $Q_i = 1$ when $A'^i_{\sigma_1^i(x),\sigma_2^i(y)} = 0$ is at most $\frac{3}{20} + o(1)$. Applying the union bound, we see that the overall error probability of our algorithm is at most

$$\frac{1}{10} + (\frac{1}{2})^{40} + \frac{3}{20} + o(1) \leq \frac{1}{3} ,$$

for large enough n. □

Since by Theorem 4, $\mathsf{Blnd}^{n,n-\frac{1}{40}n^{1-\epsilon}}_{\frac{1}{3}}$ has a communication complexity of $\Omega(n^{2-2\epsilon})$, we obtain the following result:

Corollary 12. *Every insertion-deletion n^ϵ-approximation streaming algorithm for Minimum Vertex Cover with error probability at most $\frac{1}{400}$ requires space $\Omega(n^{2-2\epsilon})$.*

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